

MSPM0_K210手写数字识别

1.K210与MSPM0通信

1.1 实验前提

本教程使用的是MSPM0G3507开发板，k210要运行**K210-AI(MSPM0G3507)** 里面的程序才能开始实验

MSPM0G *1

k210视角模块*1(要有sd卡（里面有带AI的模型）、摄像头）

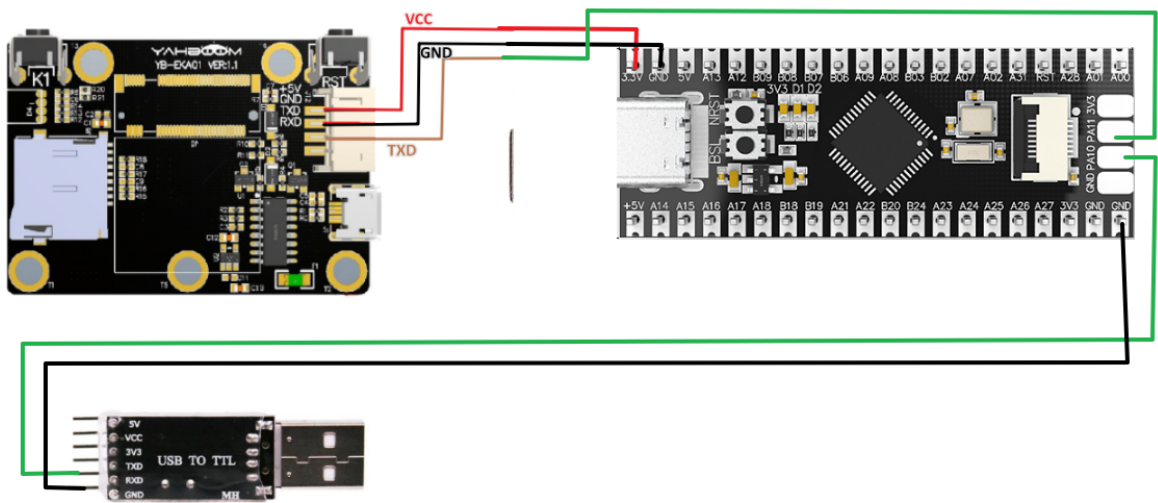
usb转ttl模块*1

1.2 实验接线

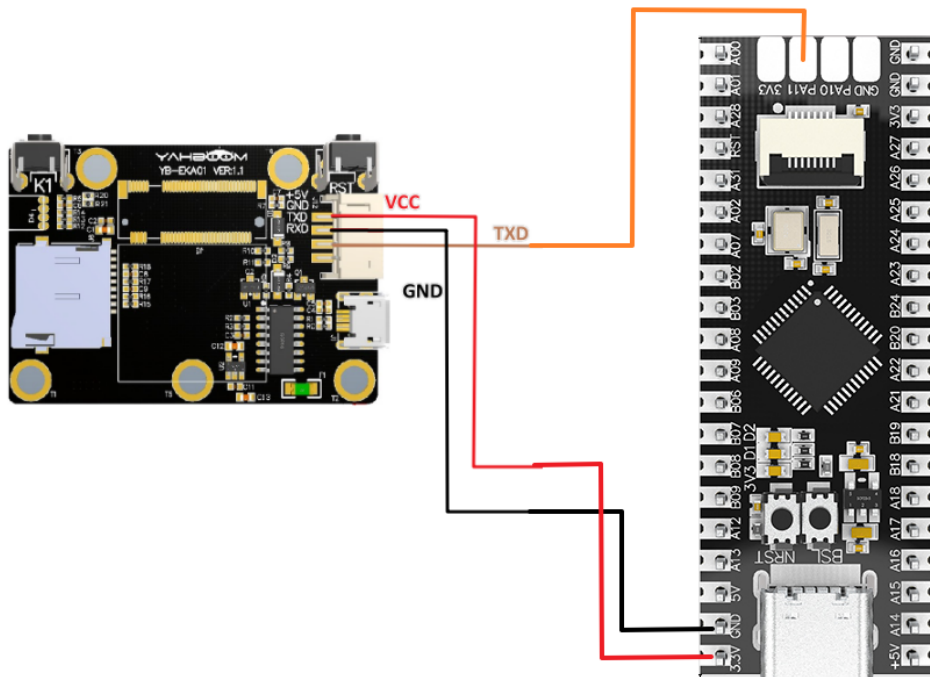
方式一：使用USB转TTL模块

MSPM0G	K210视觉模块
RX	TXD
GND	GND
VCC	5V

MSPM0G	USB转TTL模块
TX	RXD
GND	GND



方式二：使用板载Type-C口



1.3 主代码讲解

本例程将串口打印的波特率设置为115200 bps，k210模块连接的波特率设置为115200 bps。

Type Filter Text... X << < > Software > UART

PROJECT CONFIGURATION...
Project Config... 1/1 ✓ +

MSPM0 DRIVER LIBRARY ...
SYSTEM (9)
Board 1/1 ✓ +
DMA +
GPIO +
MATHACL +
Configuration NVM +
RTC +
SYSCTL 1/1 ✓ +
SYSTICK 1/1 ✓ +
WWDG +

ANALOG (6)
ADC12 +
COMP +
DAC12 +
GPAMP +
OPA +
VREF +

COMMUNICATIONS (6)
I2C +
I2C - SMBUS +
MCAN +
SPI +
UART 2/4 ✓ +
UART - LIN +

TIMERS (6)
TIMER - CAPTURE +

UART (2 of 4 Added) @

MYUART
K210_Uart

Name MYUART
Selected Peripheral UART0

Quick Profiles
UART Profiles Custom

Basic Configuration
UART Initialization Configuration

Clock Source BUSCLK
Clock Divider Divide by 1
Calculated Clock Source 32.00 MHz
Target Baud Rate 115200
Calculated Baud Rate 115211.52
Calculated Error (%) 0.01
Word Length 8 bits
Parity None
Stop Bits One
HW Flow Control Disable HW flow control

Type Filter Text...
X
<<
<
>
Software > UART

PROJECT CONFIGURATIO...

Project Config... 1/1 ✓ +

MSPM0 DRIVER LIBRARY ...

SYSTEM (9)

Board 1/1 ✓ +

DMA +

GPIO +

MATHACL +

Configuration NVM +

RTC +

SYSCCTL 1/1 ✓ +

SYSTICK 1/1 ✓ +

WWDTC +

ANALOG (6)

ADC12 +

COMP +

DAC12 +

GPAMP +

OPA +

VREF +

COMMUNICATIONS (6)

I2C +

I2C - SMBUS +

MCAN +

SPI +

UART 2/4 ✓ +

UART - LIN +

TIMERS (6)

TIMER - CAPTURE +

TIMER - COMPARE +

UART (2 of 4 Added) ②

MYUART

K210_Uart

Name

K210_Uart

Selected Peripheral

UART1

Quick Profiles

UART Profiles

Custom

Basic Configuration

UART Initialization Configuration

Clock Source

BUSCLK

Clock Divider

Divide by 1

Calculated Clock Source

32.00 MHz

Target Baud Rate

115200

Calculated Baud Rate

115211.52

Calculated Error (%)

0.01

Word Length

8 bits

Parity

None

Stop Bits

One

HW Flow Control

Disable HW flow control

```

int main(void)
{
    SYSCFG_DL_init();

    NVIC_ClearPendingIRQ(MYUART_INST_INT_IRQN); //清除串口中断标志
    NVIC_EnableIRQ(MYUART_INST_INT_IRQN); //是能串口中断

    while (1)
    {
        if (k210_msg.class_n != 0) //例程号不为空
        {
            if(k210_msg.class_n == 11) //是手写数字识别
            {
                sprintf(buff_com, "class =
%d,x=%d,y=%d,w=%d,h=%d\r\n", k210_msg.class_n, k210_msg.x, k210_msg.y, k210_msg.w, k2
10_msg.h);
                uart0_send_string(buff_com);
                k210_msg.class_n = 0;
            }
        }
        delay_ms(500);
    }
}

```

```
}
```

经过以上的程序，如果是跑这个例程，k210_msg结构体的成员就有对应的值，并通过串口打印处理

k210_msg:是接收信息的结构体，它的主要成员有

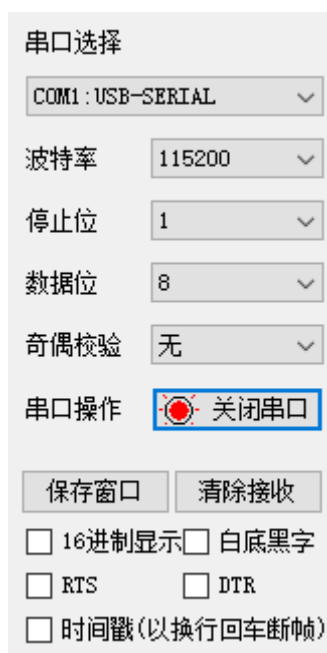
- x: 是识别出来框左上角的横坐标（范围：0-240）
- y: 是识别出来框左上角的纵坐标（范围：0-320）
- w: 是识别出来框的宽度（范围：0-240）
- h: 是识别出来框的长度（范围：0-320）
- id:是识别出来的标签
- class_n:例程编号
- msg_msg[20]:有效数据

经过数据接收的函数并处理，k210_msg的每个成员都会存储到有效信息，如果想要二次开发，直接调用就k210_msg的成员就可以了

注：必须将keil工程源码放在SDK路径下编译。

1.4 实验现象

1. 连接好线后，k210视角模块脱机运行 [k210脱机运行方法](#)
2. 串口助手设置成如图的界面



3. 然后跑手写数字识别的例程，串口助手就会打印出k210传输给MSPM0G的重要信息，如下图所示

```
id = 0  
id = 8  
id = 8  
id = 8  
id = 8  
id = 8  
id = 8  
id = 8  
id = 8  
id = 8  
id = 8  
id = 8  
id = 8  
id = 8  
id = 0  
id = 0  
id = 6  
id = 6  
id = 5  
id = 1  
id = 9  
id = 9  
id = 1  
id = 5  
id = 6  
id = 0  
id = 6  
id = 5
```

手写数字识别只传输k210_msg的id这1个成员变量。

id:是指识别出来的数字ID (0-9)